P-S-591g
December 28, 1965
SUPERSEDING
Fed. Spec. P-S-591f
March 6, 1963

FEDERAL SPECIFICATION

SOAP, LAUNDRY, BAR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of laundry soap (see 6.1).

1.2 Classification. The laundry soap covered by this specification shall be of one grade and shall be of the following sizes, as specified (see 6.2):

Size 1—1 pound bar.
Size 2—1/2 pound bar.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

QQ-S-781—Steel Strapping, Flat.
QQ-S-790—Steel Strapping, Round (Bare and Zinc-Coated).
PPP-B-636—Box, Fiberboard.
PPP-T-97—Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 536—Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.
Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547—Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Composition. The laundry soap shall be thoroughly saponified and its composi-

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tion shall conform to the requirements specified in table I when tested as specified in 4.4. The titer of the mixed fatty acids prepared from the soap shall be not lower than 29°C.

TABLE I. *Composition*

Characteristic	Minimum	Maximum
Moisture and matter volatile at 105° C., percent	—	36.0
Sum of free alkali or free acid, total matter insoluble in alcohol and sodium chloride, percent	2.0	11.0
Free alkali, calculated as sodium hydroxide (NaOH), percent	—	0.2
Free acid, calculated as oleic acid, percent	—	0.5
Matter insoluble in water, percent	—	1.0
Chloride calculated as sodium chloride (NaCl), percent	—	1.0
Rosin, percent	—	25.0
Anhydrous soap, percent	52.0	—

3.1.1 *Computation.* To determine compliance with the composition requirements of table I, the moisture and volatile matter content shall be expressed as determined on the soap and the test results of all other ingredients shall be expressed on the basis of a soap containing 36 percent moisture and volatile matter. To express these results on a 36 percent moisture and volatile matter basis, the analytical test results are multiplied by a factor F, where:

$$F = \frac{100 - 36}{100 - M}$$

M = Average percent moisture and volatile matter at 105°C. of the lot as determined from the composite.

3.2 Form. The laundry soap shall be furnished as firm bars.

3.3 Weight. The average bar weight for size 1 shall be 16 ± 1/2 ounce. The average bar weight for size 2 shall be 8 ± 1/4 ounce.[1]

[1]These requirements shall be determined on the bar per se without any moisture adjustment.

3.4 Dimensions. Size 1 laundry soap shall be 4-1/2 by 2-3/4 by 2-1/4 inches±3/8 inch in each dimension. Size 2 laundry soap shall be 4-1/2 by 2-3/4 by 1-1/8 inches ± 1/4 inch in each dimension.[1]

3.5 Color. A freshly cut surface of the laundry soap shall have a uniform color (a slight variation in color occurring within 1/4 inch of any edge shall be disregarded). The darkest portion of a freshly cut bar shall not be darker than color No. 30266 of Fed. Std. No. 595.

3.6 Odor. The laundry soap shall be free of objectionable odors at room temperature (70° to 80°F.) and in a solution of the soap in water at 130° to 140°F. An objectionable odor, for this specification, is defined as a rancid, fishy, or horsy odor.

3.7 Workmanship. The laundry soap shall be uniform in appearance and texture. Bars shall be free of foreign matter and unduly sticky surfaces. The laundry soap shall conform to the levels of quality established herein.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot shall consist of all laundry soap of the same size from one manufactured batch.

4.2.1 *Component and material inspection.* In accordance with 4.1, components and ma-

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terials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 *Examination of the end item.* The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.5. The lot size shall be expressed in units of bars of laundry soap of one size for the examinations in 4.3.2, and in units of shipping containers for the examinations in 4.3.3 and 4.3.4.

4.3.2 *Examination of laundry soap for defects in form, dimensions, color, odor, and workmanship.* The sample unit for this examination shall be one bar of soap.

Examine	Defect
Form	Not bar; not firm (crumbly).
	Not in one piece.
Dimensions	Not specified dimensions.
Color	Color not uniform.
	Darker than color chip No. 30266.
Odor	Rancid, fishy, or horsy.
Workmanship	Not uniform in appearance and texture.
	Presence of foreign matter.
	Unduly sticky.

4.3.3 *Examination for defects in average weight per bar of laundry soap.* The sample unit for this examination shall be the net contents of one shipping container. The average weight per bar shall be within the specified weight. The net contents of no less than 10 shipping containers shall be weighed. If the lot consists of less than 10 shipping containers, the net contents of all shipping containers shall be weighed.

4.3.4 *Examination of preparation for delivery.* An examination shall be made to determine that packing, weight, and markings comply with the requirements of section 5. Defects shall be scored as below. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers, fully prepared for delivery, shall be examined for closure defects.

Examine	Defect
Packing (as applicable)	Not in accordance with contract requirements.
	Arrangement or number bars of soap per shipping container not in accordance with contract requirements.
	Inadequate application of components such as: incomplete closure of case liners, container flaps; loose or inadequate strappings; bulged or distorted containers. Container not as specified; closures not accomplished by specified or required methods or materials.
	Liner material (as applicable) omitted, damaged, or not type specified.
	Any nonconforming component, component missing, damaged, or otherwise defective, affecting serviceability.
Markings (exterior and interior, as applicable)	Omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

4.3.5 *Inspection levels and AQLs for examinations.* The inspection levels, for determining the sample size, and AQLs, expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.2	I	2.5
4.3.3	S-3	—
4.3.4	S-2	2.5

4.4 Testing of the laundry soap. The methods of testing listed in table II shall be followed for each lot. Fed. Test Method Std. No. 536 shall be used as indicated in the table. The lot size shall be expressed in pounds of laundry soap. The sample unit for testing shall be a bar of laundry soap. A one-pound composite shall be prepared from the sample units randomly selected at the frequency listed below. The cutting, grinding and mixing for the preparation of the composite sample shall be conducted in

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an atmosphere of relative humidity within 5 percent of the relative humidity prevailing at time and place of weighing. The prepared composite shall be immediately placed in a clean, dry, airtight container marked with identifying lot number and date. Care shall be taken to prevent contamination or alteration of the composite during sampling, grinding, weighing, and storing. Duplicate determinations shall be made on the composite. The average of the two determinations shall comply with the requirements.

Lot size (pounds)	Sample frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and application of level of packing shall be in accordance with Fed. Std. No. 102.

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

TABLE II. *Instructions for testing of the end item*

Characteristic	Specification reference		Requirements applicable to comp. samp.	Number of determinations	Results reported as
	Requirement	Test method[1]			
Moisture and matter volatile at 105° C.	Table I	201	X	2	Average of 2 determinations to nearest 0.1 percent.
Sum of free alkali or free acid, total matter insoluble in alcohol, and sodium chloride	Table I	301	X	2	Average of 2 determinations to nearest 0.1 percent.
Free alkali calculated as sodium hydroxide (NaOH)	Table I	302	X	2	Average of 2 determinations to nearest 0.1 percent.
Chloride calculated as sodium chloride (NaCl)	Table I	501	X	2	Average of 2 determinations to nearest 0.1 percent.
Matter insoluble in water	Table I	303	X	2	Average of 2 determinations to nearest 0.1 percent.
Anhydrous soap	Table I	401	X	2	Average of 2 determinations to nearest 0.1 percent.
Free acid, calculated as oleic acid	Table I	302	X	2	Average of 2 determinations to nearest 0.1 percent.
Rosin (quantitative test)	Table I	701	X	2	Average of 2 determinations to nearest 0.1 percent.
Titer mixed fatty acids prepared from the soaps	3.1	901	X	2	Average of 2 determinations to nearest 1° C.

[1]All tests are performed in accordance with Fed. Test Method Std. No. 536.

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5.1.1 *Level A.* Sixty size 1 or one hundred and twenty size 2 unwrapped bars of laundry soap shall be packed on end in a snug-fitting fiberboard shipping container conforming to style RSC, V3c or V3s of PPP-B-636. Each shipping container shall be provided with a snug-fitting top and bottom pad made of the same material as the container. Size 1 bars of soap shall be arranged in 2 layers with 5 rows in each layer and 6 bars of soap per row. Size 2 bars of soap shall be arranged in 2 layers with 5 rows in each layer and 12 bars of soap per row. Each shipping container shall be provided with a case liner constructed in accordance with type I of MIL-L-10547, except that the material shall consist of a lamination of 30-pound basis weight (24 by 36—500) kraft paper and 0.001-inch polyethylene. The bottom and side seams shall be heat sealed with seams not less than 3/8 inch in width. Closure of the case liner shall be accomplished by heat sealing or by a lock-seam, folded flat against the contents. Excess air shall be expelled prior to heat sealing of the closure.

5.1.1.1 *Strapping.* Packed and sealed fiberboard shipping containers shall be strapped with flat steel, twist tie-wire, round wire strapping, or filament reinforced pressure-sensitive adhesive tape as specified (see 6.2). The flat steel or twist tie-wire shall conform to type I or III, class B of QQ-S-781. The type I strapping shall be not less than 3/8 by 0.015 inch. Round wire strapping shall be not less than 0.0915 inch in diameter conforming to finish 2 of QQ-S-790. The filament reinforced pressure-sensitive adhesive tape shall be not less than one-half inch in width conforming to type IV of PPP-T-97. V3c corrugated fiberboard containers shall be strapped with flat steel strapping or tape as specified herein. V3s solid fiberboard containers shall be strapped with either flat steel, round, twist tie-wire strapping, or tape as specified herein. Steel strapping or tape shall be applied at right angles to each other with one band centered over the top, sides, and bottom and the other band centered over the top, ends, and bottom. Tape shall be overlapped on itself a minimum of eight inches.

5.1.2 *Level B.* Unless otherwise specified, sixty size 1 or one hundred and twenty size 2 unwrapped bars of soap shall be packed on end in a snug-fitting fiberboard shipping container conforming to style RSC, type CF or SF, class domestic, grade 200 of PPP-B-636, except the manufacturer's joint shall be lapped and stitched. Each shipping container shall be provided with a snug-fitting top and bottom pad made of the same material as the container and a case liner as specified in 5.1.1. Size 1 bars of soap shall be arranged in 2 layers with 5 rows in each layer and 6 bars of soap per row. Size 2 bars of soap shall be arranged in 2 layers with 5 rows in each layer and 12 bars of soap per row. Closure shall be in accordance with method II of the appendix of the container specification. When specified (see 6.2), the shipping container shall conform to V3c or V3s of PPP-B-636.

5.1.3 *Level C.* The bars of soap shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military requirements.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129 and shall contain one of the following legends, as applicable:

"1 CASE
(60-1 lb. bars)."
or
"1 CASE
(120-1/2 lb. bars)."

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6. NOTES

6.1 Intended use. The laundry soap covered by this specification is intended for use with soft to moderately hard water for hand dishwashing, cleaning of hard surfaces, and hand laundering where alkali salts are desirable.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Size required (see 1.2).
(*c*) Selection of the applicable level of packing (see 5.1).
(*d*) Whether steel strapping or tape shall be required (see 5.1.1.1).
(*e*) Whether waterproof containers are required for level B shipments (see 5.1.2).

6.3 Basis of purchase. Unless otherwise specified, sampling of the soap shall be performed at manufacturer's plant. Deliveries of soap containing more than 36 percent moisture and matter volatile at 105°C. will be rejected without further tests.

6.3.1 The soap should be purchased by net weight, provided the moisture and matter volatile at 105°C. is neither above nor below 34 percent.

6.3.2 On deliveries containing 27 to 36 percent moisture and matter volatile at 105°C., payment should be calculated on the basis of a product containing 34 percent moisture, that is, 0.66 pound nonvolatile matter should be considered 1 pound of soap. On deliveries of soap containing less than 27 percent moisture and matter volatile at 105°C., settlement should be made as though the soap contained 27 percent moisture and matter volatile. The net weight of the material to be paid for on a 34 percent moisture and matter volatile basis, is calculated as follows:

$$W = \frac{R \times (100 - L)}{66}$$

where:

W = net weight of materials to be paid for on a 34 percent moisture and matter volatile basis.
R = net weight of material as received.
L = percentage of loss at 105° C.

Example for more than 34 percent moisture and volatile matter

where:

$$W = \frac{1 \times (100 - 35)}{66} = 0.985 \text{ pound}$$

Cost of soap = number of pounds of soap as received × 0.985 × price per pound.

Example for less than 34 percent moisture and volatile matter

where:

$$W = \frac{1 \times (100 - 31)}{66} = 1.045 \text{ pounds}$$

Cost of soap = number of pounds of soap as received × 1.045 × price per pound.

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Soap, not otherwise indexed by name, other than liquid.
Carload minimum weight 36,000 pounds.

Motor:

Soap, not otherwise indexed, other than liquid.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

P-S-591f; Soap, Laundry Bar Proposed Revision of Project No. 7930-0112

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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